

JEAN-MILAN ALBAREDE

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EDUCATION

Purdue University, College of Engineering, West Lafayette, Indiana

Bachelor's degree, Electrical and Electronics Engineering

May 2028

GPA: 4.00

Relevant Coursework

Differential Equations, Linear Algebra, EE Fundamentals I, Intro to Digital System Design, Physics I & II, Advanced C

SKILLS

Software: Altium Designer, LTspice, Git/GitHub, Linux, Microsoft Excel, PowerPoint, SolidWorks, Fusion 360

Programming: Python, C, Verilog, Java, MATLAB, LaTeX

Language: Fluent in Russian, French, and English

Other: Data analysis w/ Python, Soldering, Oscilloscope, Signal Generator, I2C, UART, Self-Motivated

PROFESSIONAL EXPERIENCE

N6 Tec, Biotech Startup - Pleasanton, CA

Summer 2023

Engineering Intern

- Created part models in **SolidWorks**, applying **GD&T** tolerances; received feedback through iterative design reviews and markups.
- Conducted **parts documentation** by compiling component data from **McMaster-Carr** into company records.
- Prototyped and fabricated small components that supported company projects utilizing additive manufacturing.

ENGINEERING ORGANIZATIONS

Embedded Systems @ Purdue, Harmonicore Project (FPGA Subteam)

Jan. 2026 - Present

- Converted Python audio-processing algorithms into **RTL** diagrams and **Verilog** for an **Artix-7 FPGA**, implementing live effects like echo, chorus, or reverb on analog audio input.
- Developed and verified **test benches** in **Xilinx Vivado** to ensure correct real-time audio effects, leveraging BRAM, FIFO, and lookup tables.

IEEE MTT-S, Microwave Theory & Techniques Society (Member)

Sep. 2025 - Dec. 2025

- Aided in designing the transceiver's **signal chain** for both transmit (Tx) and receive (Rx) paths, applying knowledge of **antenna theory** and **RF architecture**.
- Evaluated RF front-end components (PLLs, LNAs, Mixers, Phase Shifters) to optimize **noise figure** and **insertion loss**.

TECHNICAL PROJECTS

Wi-Fi Propagation & Path Loss Simulator

- Built a physics-based **ray-tracing** simulation algorithm using **NumPy** to model wireless signal decay via the Log-Distance Path Loss model, accounting for frequency bands (2.4/5GHz) and environmental clutter.
- Utilized **Matplotlib**, enabling users to design custom floor plans and visualize signal coverage through heatmaps.

Your Body Guardian – Autonomous Pill Dispensing System

- **Collaborated** in a **four-person team** to design and prototype a **microcontroller-based** pill dispensing system addressing medication non-adherence, optimizing usability, reliability, and cost under tight time constraints.
- Earned **1st place** for final **project presentation**, chosen from a class of **~100** for technical execution & communication.

Analog Temperature Sensor Conditioning PCB

- Designed and simulated (**Altium/LTspice**) a complete analog signal-conditioning PCB using a TMP36 sensor, incorporating a 5V→3.3V LDO, op-amp gain stage, RC low-pass filtering, and unity-gain buffering.

Digital Picture Frame Project

- Repurposed a laptop display into a custom digital picture frame using a Raspberry Pi and an LCD driver board.
- Developed a Python-based GUI that integrates Flickr's API to automatically fetch and display photos, utilizing flickrapi, Pillow, and tkinter.

Anti-Vaping Support Device

- Designed and prototyped a portable, **battery-powered device** that mimics the inhalation sensation of vaping using an ultrasonic atomizer, a voltage regulation board, and a control board.